



SIMATS ENGINEERING

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Engineer to Excel

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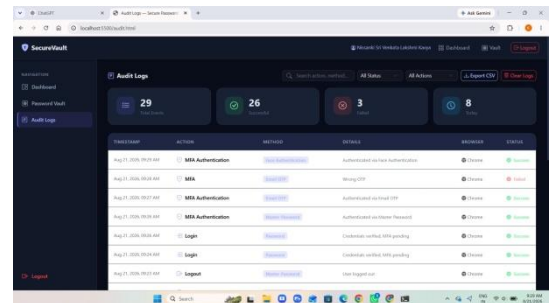
Course Code: CSA5115

Slot: A

Course Name: Cryptography and Network Systems

Course Faculty: Dr. T Sangeetha

Project Title: Secure Password Manager with Multi-Factor Authentication



Review

Module

Project Description

The Secure Password Manager with Multi-Factor Authentication (MFA) is a cybersecurity application designed to securely store, manage, and protect users' passwords and sensitive login credentials. Instead of remembering multiple passwords or storing them insecurely, users can maintain their credentials in an encrypted password vault that can only be accessed after successful authentication.

The system uses multi-factor authentication to provide an additional layer of security beyond a username and password. After entering the primary password, the user must complete a second authentication step, such as an OTP, authentication code, or email verification, before gaining access to the password vault.

Passwords stored in the vault are protected using strong encryption and cryptographic hashing techniques. The application can also include features such as password generation, password-strength checking, secure password retrieval, credential search, automatic session timeout, and failed-login detection.

The main objective of the project is to demonstrate how encryption, secure authentication, hashing, access control, and MFA can be combined to build a practical password-management system. The project helps reduce risks associated with weak, reused, or exposed passwords while demonstrating important concepts in information and network security.

Student Signature

Guide Signature